Coronary Artery Disease Bayesian Classification: Derivation, Implementation, and Empirical Validation on the UCI Cleveland Dataset

***Abstract*—We investigate the classification behavior and probability calibration of a hybrid Naive Bayes model on the canonical UCI Cleveland Clinic Foundation cohort. Rather than forcing all variables into an identical distributional family or relying on heuristic binning, we combine Gaussian likelihoods for four continuous clinical measurements with Laplace-smoothed categorical likelihoods for two discrete symptom indicators. An audit across all 13 predictor columns confirmed that no exact duplicate rows exist, and all 303 patient records were retained. Ground-truth angiography narrows down the 303 cases into 164 controls and 139 coronary artery disease (CAD) cases. Using a leak-free protocol with repeated stratified 10×5-fold cross-validation and a held-out test split, incorporating symptom categories improved mean cross-validation ROC-AUC from 0.732 ± 0.052 to 0.831 ± 0.045, a gain of 0.099. Discrimination closely matched ℓ_2-regularized logistic regression (0.831 ± 0.045 vs. 0.830 ± 0.049; paired-fold Wilcoxon $p = 0.652$). On the 80/20 holdout partition, the mixed classifier reached 80.3% accuracy [95% CI: 70.5%–90.2%], 82.1% recall, and 0.891 ROC-AUC [95% CI: 0.803–0.960]. However, while the model demonstrated competitive ranking ability, probability calibration was inferior to logistic regression (Brier score 0.183 vs. 0.168), driven in part by within-class predictor associations such as an inverse correlation between age and peak heart rate ($r = -0.523$ in controls). These empirical results characterize Bayesian classification trade-offs on a retrospective registry and do not constitute evidence for clinical decision support.**

***Index Terms*—Bayesian classification, Mixed Naive Bayes, coronary artery disease, Laplace smoothing, Gaussian likelihood, UCI Cleveland dataset, model calibration, diagnostic machine learning.**

I. Introduction

Coronary artery disease (CAD) stems from the progressive accumulation of atherosclerotic plaque within the epicardial coronary vessels and remains a principal cause of cardiovascular mortality globally [2]. While invasive coronary angiography serves as the diagnostic standard for measuring luminal obstruction, the catheterization procedure carries non-trivial risks of vascular and systemic complications [3]. Consequently, statistical and machine learning algorithms that analyze non-invasive physiological signs and reported symptoms have attracted sustained research interest. However, retroactively fitting statistical classifiers to historic registries serves an exploratory engineering objective; it does not demonstrate readiness for prospective patient management.

For modest sample sizes, Naive Bayes provides an attractive baseline due to its transparency: prior class probabilities and likelihood parameters have closed-form estimators and clear probabilistic meanings. In clinical settings, patient profiles are inherently heterogeneous. Indicators like resting arterial pressure, peak exercise heart rate, and total serum cholesterol are continuous quantities with physical units, whereas presentation symptoms like chest-pain character and exercise-induced discomfort are discrete categories. Enforcing a single Gaussian family across all attributes assumes an artificial metric distance between unordered categories, while discretizing continuous variables into subjective intervals discards fine-grained measurement variance.

In this work, we implement a hybrid Naive Bayes classifier that pairs parametric Gaussian densities for continuous variables with add-one Laplace-smoothed tables for categorical symptoms. In preparing the cohort, we resolved conflicting target encodings in redistributed copies and verified that no exact duplicate rows exist across the 13 predictor fields, retaining all 303 records. To protect against subtle data leakage, all parameter estimation and feature scalings were confined strictly to training folds. We benchmark the resulting mixed-type model against an ablated continuous-only Gaussian baseline and an ℓ_2-regularized logistic regression. In addition to conventional discrimination metrics across repeated cross-validation and an independent holdout, we evaluate model calibration through the Brier score and examine empirical departures from the underlying conditional independence assumptions.

II. Problem Statement

Our objective is to evaluate whether a hybrid Naive Bayes architecture can accurately separate patients with angiographically confirmed CAD from disease-free controls using six selected clinical features. The target variable reflects coronary stenosis measured at the time of catheterization; because the dataset provides no prospective follow-up or longitudinal survival times, this task is strictly a cross-sectional, retrospective classification problem rather than a long-term risk prognosis.

We address three specific methodological questions:

1) *Ablation of Symptom Information:* How much discriminative gain does the addition of discrete symptom categories provide over an ablated model relying exclusively on continuous physiological measurements?
2) *Discriminative vs. Generative Baseline:* How does the ranking performance and probability calibration of this generative mixed-type classifier compare against standard regularized logistic regression?
3) *Robustness to Violated Assumptions:* How sensitive is the model to demonstrable violations of normality and

class-conditional independence in real-world clinical measurements?

All analyses reflect retrospective performance on this specific historic cohort; they do not establish transportability to modern patient populations or prospective clinical efficacy.

III. Literature Review

The Cleveland Clinic Foundation heart disease cohort was originally compiled by Detrano *et al.* [4] as part of an international initiative across four clinical institutions (Cleveland, Hungary, Zurich, and Long Beach) to derive a probability-based diagnostic algorithm. Distributed through the UCI Machine Learning Repository [1], the Cleveland portion became an established benchmark in machine learning literature. Most benchmark studies simplify the clinical target from Detrano's five-level diagnostic coding into a binary classification task distinguishing disease absence ($\texttt{num} = 0$) from disease presence ($\texttt{num} \in \{1, 2, 3, 4\}$).

The theoretical properties of Naive Bayes classifiers under violated conditional independence have been analyzed extensively. Domingos and Pazzani [5] demonstrated that zero-one classification error is governed by the sign of the posterior odds ratio rather than the numerical accuracy of individual joint probability estimates. As a result, Naive Bayes can achieve competitive classification accuracy even when the conditional independence assumption is heavily violated. Rish [6] reached similar conclusions across synthetic and empirical benchmarks, noting that classification ranking frequently remains robust. However, as shown by Niculescu-Mizil and Caruana [10], competitive discrimination does not imply well-calibrated posterior probabilities; Naive Bayes often yields push-to-boundary overconfident probabilities. Assessing model quality therefore requires evaluating both threshold-free ranking metrics (ROC-AUC) [8] and proper scoring rules such as the Brier score [9].

Extensive comparative studies on the Cleveland dataset illustrate different algorithmic trade-offs. Khanna *et al.* [13] benchmarked support vector machines, logistic regression, and artificial neural networks, finding that kernel and neural approaches achieved high cross-validation discrimination at the expense of model transparency. Amin *et al.* [14] evaluated seven distinct learning algorithms and emphasized that feature selection substantially affects predictive stability. Similarly, Akella and Akella [15] explored multiple tree-based and neural architectures to establish open-source performance baselines. Focusing on Bayesian methods, Meidina and Abidin [16] applied particle swarm optimization and gain-ratio selection to Naive Bayes on the 303 Cleveland records. While these studies report various accuracy metrics, differing validation protocols and data preprocessing choices complicate direct numerical comparisons with our findings.

IV. Dataset and Problem Formulation

A. Dataset Provenance and Target Ground Truth

The UCI Heart Disease collection contains patient records from four distinct clinical centers compiled by Robert Detrano, M.D., Ph.D., and collaborators [1], [11]. Among these, the Cleveland Clinic Foundation dataset is the most widely cited benchmark. While the source repository catalogs 76 clinical attributes, empirical studies routinely focus on a 14-variable curated subset.

Data integrity audits revealed two important preprocessing pitfalls. First, several popular redistributed versions (frequently circulated as `heart.csv`) invert the outcome variable, labeling healthy controls as 1 and CAD patients as 0, or omit variable documentation. Second, the repository documentation by Aha [11] notes that the unformatted file `cleveland.data` contained file corruptions, making `processed.cleveland.data` the authoritative source for the 14-attribute representation. In this canonical file, the target attribute `num` represents the angiographic diagnosis of heart disease, where 0 indicates absence (angiographic disease-free status) and values 1 through 4 indicate presence under the dataset's coding. The number of major vessels visualised by fluoroscopy is recorded separately by the predictor variable `ca`, not `num`.

We define the binary classification target as:

$$C = \mathbb{I}(\texttt{num} > 0) \tag{1}$$

where $C = 0$ represents non-significant vessel disease ($< 50\%$ diameter reduction, $N_0 = 164$) and $C = 1$ represents clinically significant stenosis in one or more vessels ($N_1 = 139$), for a total of $N = 303$ patients. Furthermore, a systematic audit across all 13 predictor fields identified no exact duplicate rows; all 303 records are therefore retained without ad-hoc record removal.

B. Feature Selection and Clinical Representation

To systematically analyze the interaction between continuous physiological variables and discrete symptoms in a hybrid Bayesian setting, we selected six clinically standard attributes from the intake examination:

1) **Age** (`age`, years): patient age in completed years.
2) **Resting blood pressure** (`trestbps`, mm Hg): systolic blood pressure measured at clinical intake.
3) **Serum cholesterol** (`chol`, mg/dl): total fasting serum cholesterol.
4) **Maximum heart rate** (`thalach`, beats/min): peak heart rate attained during treadmill exercise testing.
5) **Chest pain type** ($\texttt{cp} \in \{1, 2, 3, 4\}$): categorical classification indicating 1 for typical angina, 2 for atypical angina, 3 for non-anginal pain, and 4 for asymptomatic presentation.
6) **Exercise-induced angina** ($\texttt{exang} \in \{0, 1\}$): binary symptom indicator (0 absent, 1 present during exertion).

These six attributes are fully recorded across all 303 patients, avoiding the need for synthetic data imputation (in contrast to fluoroscopy vessel count `ca`, missing in 4 records, and thallium scintigraphy `thal`, missing in 2 records). Discrete symptom codes are treated as unordered nominal categories rather than continuous linear values, avoiding the imposition of false geometric distances between distinct pain sensations.

TABLE I
COHORT BALANCE: HEALTHY ($C = 0$, $N = 164$), CAD ($C = 1$, $N = 139$). CONTINUOUS ENTRIES ARE MEAN ± SD; `cp` ENTRIES GIVE COUNTS FOR LEVELS 1–4 IN ORDER.

Attr.	Meaning	Model	Healthy (C=0)	CAD (C=1)
age	Age (years)	Gauss. MLE	52.59 ± 9.51	56.63 ± 7.94
trestbps	Resting BP (mm Hg)	Gauss. MLE	129.25 ± 16.20	134.57 ± 18.77
chol	Cholesterol (mg/dl)	Gauss. MLE	242.64 ± 53.46	251.47 ± 49.49
thalach	Max HR (bpm)	Gauss. MLE	158.38 ± 19.20	139.26 ± 22.59
cp	Chest pain (4 levels)	Laplace cat.	$16, 41, 68, 39$	$7, 9, 18, 105$
exang	Exercise angina	Laplace cat.	86.0%/14.0%	45.3%/54.7%

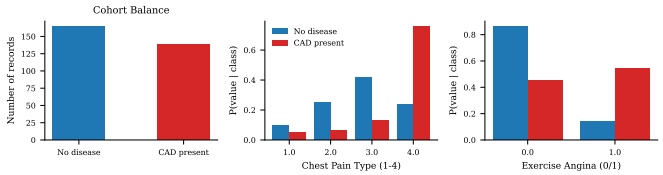


Fig. 1. Class counts and the within-class distributions of chest-pain type (`cp`) and exercise-induced angina (`exang`). The CAD group contains more asymptomatic chest-pain cases (`cp` = 4) and more exercise-provoked angina.

Table I outlines the demographic and clinical characteristics of the cohort by disease status.

Figure 1 illustrates the noticeable distributional contrasts between the two classes across both symptom indicators: CAD patients exhibit a high frequency of asymptomatic presentations (`cp` = 4) and exercise-provoked angina.

V. MATHEMATICAL FOUNDATIONS

A. Bayesian Decision Theory and Conditional Independence

Let $\mathbf{x} = (x_1, \ldots, x_d)^\top$ represent the vector of observed clinical features, and let $C \in \{0, 1\}$ denote the binary disease indicator. By Bayes' theorem, the posterior class probability given an observation is expressed as:

$$P(C = c \mid \mathbf{x}) = \frac{P(C = c)\, P(\mathbf{x} \mid C = c)}{\sum_{k \in \{0,1\}} P(C = k)\, P(\mathbf{x} \mid C = k)}. \tag{2}$$

Here, the prior probability $P(C = c)$ reflects the prevalence of class c in the training partition, while $P(\mathbf{x} \mid C = c)$ is the class-conditional likelihood of the observed profile.

The Naive Bayes formulation simplifies this likelihood by assuming that all features are conditionally independent given the true class C:

$$P(\mathbf{x} \mid C = c) = \prod_{j=1}^{d} P(x_j \mid C = c). \tag{3}$$

To prevent numerical underflow when computing products of small probabilities in floating-point operations, calculations are performed on the logarithmic scale. The unnormalized log-score for class c is:

$$\begin{aligned} \log S(c \mid \mathbf{x}) = \log P(C = c) \\ + \sum_{j \in \mathrm{NUM}} \log f_j(x_j \mid C = c) \\ + \sum_{j \in \mathrm{CAT}} \log P(x_j \mid C = c), \end{aligned} \tag{4}$$

where $\mathrm{NUM} = \{\texttt{age}, \texttt{trestbps}, \texttt{chol}, \texttt{thalach}\}$ indexes the continuous measurements and $\mathrm{CAT} = \{\mathrm{cp}, \mathrm{exang}\}$ denotes the categorical symptom variables.

A patient observation is assigned to the CAD-positive class whenever the difference in unnormalized log-scores meets or exceeds a log-odds threshold τ:

$$\hat{y} = \begin{cases} 1, & \log S(1 \mid \mathbf{x}) - \log S(0 \mid \mathbf{x}) \geq \tau, \\ 0, & \text{otherwise}, \end{cases} \tag{5}$$

where the log-odds threshold is defined as $\tau = \log\left(\frac{\theta}{1-\theta}\right)$ for a chosen operating probability threshold $\theta \in (0, 1)$. The standard Bayesian maximum a posteriori rule uses $\theta = 0.5$, which yields $\tau = 0$. Adjusting θ alters the operating sensitivity and specificity without changing the underlying likelihood models.

B. Continuous Features: Gaussian Likelihood

For each continuous variable $x_j \in \mathrm{NUM}$, we model the class-conditional distribution as a univariate Gaussian density:

$$f_j(x_j \mid C = c) = \frac{1}{\sqrt{2\pi\sigma_{jc}^2}} \exp\Big(-\frac{(x_j - \mu_{jc})^2}{2\sigma_{jc}^2}\Big). \tag{6}$$

For a subset of N_c training observations belonging to class c indexed by I_c, parameters are estimated via maximum likelihood:

$$\mu_{jc} = \frac{1}{N_c} \sum_{i \in I_c} x_{ij}, \tag{7}$$

$$\sigma_{jc}^2 = \max\Big(\frac{1}{N_c} \sum_{i \in I_c} (x_{ij} - \mu_{jc})^2,\, \varepsilon_{\text{floor}}\Big). \tag{8}$$

The variance uses the maximum likelihood denominator N_c ($\mathrm{ddof} = 0$), bounded below by a variance floor $\varepsilon_{\text{floor}} = 10^{-6}$ to prevent numerical division by zero.

C. Categorical Features: Laplace Smoothing

For each discrete symptom variable $x_j \in \mathrm{CAT}$ taking values in a discrete category set $\mathcal{K}_j$, empirical category frequencies are smoothed using Laplace's rule of succession (add-one smoothing):

$$P(x_j = v \mid C = c) = \frac{N_{jc}(v) + \alpha}{N_c + \alpha\, |\mathcal{K}_j|}, \tag{9}$$

where $N_{jc}(v)$ is the count of training observations in class c displaying category v. We set $\alpha = 1$, with cardinalities $|\mathcal{K}_{\mathrm{cp}}| = 4$ and $|\mathcal{K}_{\mathrm{exang}}| = 2$. This smoothing guarantees that unseen or rare symptom categories in a training fold are assigned a strictly positive probability, preventing zero-likelihood annihilation of the total class score.

D. Log-Sum-Exp Normalization

Posterior probabilities are calculated using the numerically stable log-sum-exp transformation. Letting $M = \max_{c \in \{0,1\}} \log S(c \mid \mathbf{x})$, we compute:

$$P(C = 1 \mid \mathbf{x}) = \frac{\mathrm{e}^{\log S(1|\mathbf{x}) - M}}{\mathrm{e}^{\log S(0|\mathbf{x}) - M} + \mathrm{e}^{\log S(1|\mathbf{x}) - M}}. \tag{10}$$

VI. Methodology and Experimental Design

A. Leakage-Free Preprocessing

To strictly eliminate train-test data leakage, all statistical parameters—including class priors $P(C = c) = N_c/N_{\text{train}}$, Gaussian moments $(\mu_{jc}, \sigma^2_{jc})$, and categorical frequency tables—are computed exclusively on the designated training partition within each cross-validation fold or holdout split. For baseline models, feature standardization and one-hot encoding pipelines are fitted solely on training folds before being applied to held-out test data. Hyperparameters ($\alpha = 1.0$, $\varepsilon_{\text{floor}} = 10^{-6}$) were fixed *a priori* rather than tuned on test evaluations.

B. Validation Protocols

Evaluation employs two complementary validation structures:

1) **Stratified 80/20 Holdout:** An independent holdout test split (random seed 42) assigns 242 records to the training set (131 controls, 111 CAD cases) and 61 records to testing (33 controls, 28 CAD cases). Percentile 95% confidence intervals are derived from 1,000 non-parametric bootstrap resamples of the test set.
2) **Repeated Stratified 10×5-fold Cross-Validation:** To mitigate partition variance in a modest sample of 303 observations, five-fold stratified cross-validation is repeated ten times with distinct random shuffles (seed 1), generating 50 individual test folds.

Within each repetition, each patient record receives exactly one out-of-fold prediction. The reported cross-validation performance represents the arithmetic mean and sample standard deviation across all 50 folds.

C. Baselines

We compare the proposed hybrid model against three benchmark models:

1) **Gaussian NB (Numeric Only):** An ablated Naive Bayes classifier fitted exclusively on the four continuous features (`age`, `trestbps`, `chol`, `thalach`) to measure the direct contribution of symptom categories.
2) **ℓ_2-Regularized Logistic Regression:** A discriminative linear baseline trained on standardized continuous variables and one-hot encoded categorical variables ($C = 1.0$, maximum iterations 2,000).
3) **Decision Tree (Fixed Depth):** A non-parametric baseline with fixed hyperparameters (maximum depth 3, minimum leaf size 5) evaluated without post-hoc tree tuning.

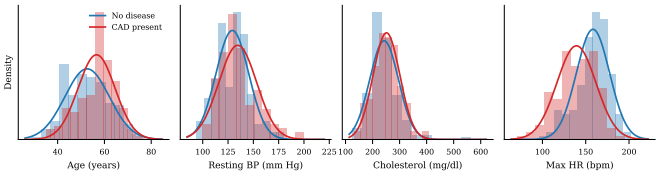


Fig. 2. Within-class histograms of the continuous predictors with fitted Gaussian curves. Maximum heart rate (`thalach`) separates the two classes most clearly.

D. Metrics

Model performance is quantified across discrimination, classification balance, and probability calibration:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}, \tag{11}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{12}$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}, \qquad F_1 = 2\frac{\text{Precision}\,\text{Recall}}{\text{Precision} + \text{Recall}}. \tag{13}$$

Discrimination across all possible operational cutoffs is summarized by the area under the receiver operating characteristic curve (ROC-AUC) [8]. Probability calibration is evaluated using the Brier score [9]:

$$\text{BS} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2, \tag{14}$$

which measures the mean squared difference between predicted probabilities p_i and true binary labels $y_i \in \{0, 1\}$. Paired-fold AUC differences between Mixed NB and logistic regression across all 50 folds are tested using the non-parametric Wilcoxon signed-rank test. Computations use scikit-learn [7] and SciPy.

VII. Results

A. Assumption Diagnostics

We examined the validity of the Gaussian likelihood assumption by conducting within-class Shapiro–Wilk normality tests on all continuous features. At the nominal $\alpha = 0.05$ level, the assumption of normality is formally rejected for resting blood pressure in CAD patients ($p = 1.17 \times 10^{-4}$), serum cholesterol in controls ($p = 2.88 \times 10^{-9}$), and peak heart rate in controls ($p = 5.43 \times 10^{-4}$). Serum cholesterol in particular displays substantial positive skewness (skew $= 1.13$) with several extreme readings exceeding 400 mg/dl. Within the CAD group, age also deviates from normality ($p = 0.0023$), as does resting blood pressure in controls ($p = 0.0116$). Conversely, the tests do not reject normality for cholesterol in CAD records ($p = 0.403$) or peak heart rate in CAD records ($p = 0.352$). Figure 2 shows the empirical histograms alongside the fitted class-conditional Gaussian distributions.

Pairwise Pearson correlation analyses (Fig. 3) indicate that while most feature pairs show weak associations ($|r| < 0.25$), a notable negative correlation exists between age and peak

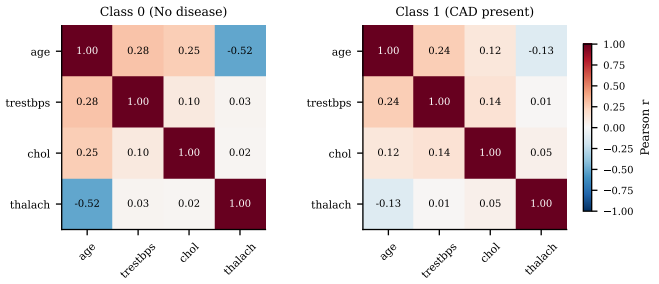


Fig. 3. Pearson correlation matrices calculated separately by class. Among controls, the largest association is between age and `thalach` ($r = -0.52$).

exercise heart rate among controls ($r = -0.523$). This demonstrates a clear violation of the class-conditional independence assumption in real physiological data.

B. Observed Feature Patterns

Inspection of marginal distributions in Table I reveals key clinical trends: relative to controls, CAD patients are older by an average of 4.04 years, exhibit higher resting blood pressure by 5.32 mm Hg, and have higher serum cholesterol by 8.83 mg/dl, while their maximum exercise heart rate is lower by 19.12 beats/min.

Marked separation appears in the categorical symptom variables. Asymptomatic chest pain (`cp` $= 4$) was reported in 75.5% of CAD patients (105/139) compared to only 23.8% of controls (39/164). Similarly, exercise-induced angina occurred in 54.7% of CAD cases (76/139) versus 14.0% of controls (23/164). These substantial distributional differences provide the underlying mechanism for the performance improvement observed when symptom categories are included in the model.

C. Predictive Performance and Ablation

Table II reports performance across both the repeated cross-validation folds and the holdout split. In the repeated 10×5-fold cross-validation, the proposed Mixed NB model achieved an average accuracy of $74.8\% \pm 4.8\%$ and an ROC-AUC of 0.831 ± 0.045. In the continuous-only Gaussian NB ablation, ROC-AUC fell by 0.099 to 0.732 ± 0.052, while accuracy dropped by 7.1 percentage points to $67.7\% \pm 5.0\%$ and recall dropped by 10.7 percentage points ($60.7\% \pm 8.2\%$ vs. $71.4\% \pm 7.3\%$).

Comparing Mixed NB with regularized logistic regression across the 50 folds showed nearly identical discrimination: the average fold-level AUC difference was $+0.0011 \pm 0.0188$ in favor of Mixed NB, which was not statistically significant on an exploratory paired Wilcoxon signed-rank test ($p = 0.652$). Recall was slightly higher for Mixed NB (71.4% vs. 70.7%), while specificity and precision favored logistic regression (78.8% vs. 77.6% and 74.3% vs. 73.3%). Figure 4 visualizes the metric comparisons across cross-validation folds.

D. Holdout Classification

On the independent test set ($N = 61$, 28 CAD cases, 33 controls), Mixed NB correctly identified 23 true positives and

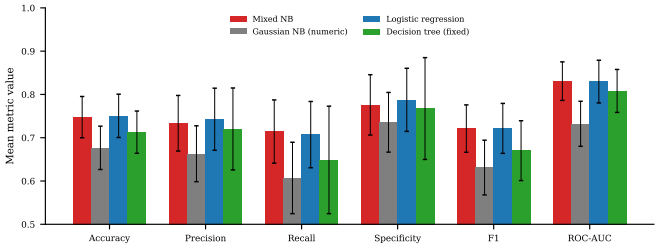


Fig. 4. Average values across the 50 validation folds; error bars show one standard deviation. Mixed NB and logistic regression perform similarly, while the numeric-only Gaussian model performs worse.

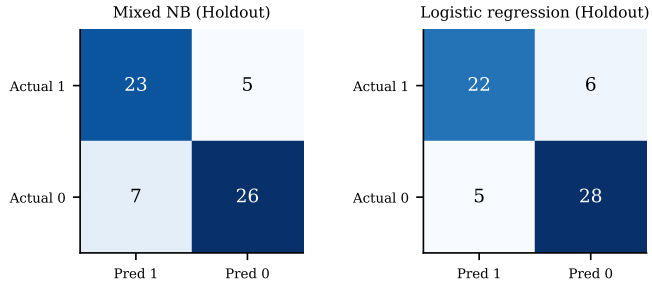


Fig. 5. Confusion matrices for the stratified test split ($N = 61$), with CAD treated as the positive class. For Mixed NB, the counts are 23 TP, 5 FN, 7 FP, and 26 TN.

26 true negatives, yielding 80.3% accuracy [95% CI: 70.5%–90.2%], 82.1% recall, 78.8% specificity, and 0.891 ROC-AUC [95% CI: 0.803–0.960] (Fig. 5). Logistic regression attained 82.0% accuracy [95% CI: 72.1%–90.2%], 78.6% recall, 84.8% specificity, and 0.898 ROC-AUC [95% CI: 0.814–0.966]. The broad, overlapping bootstrap confidence intervals reflect sample size constraints and show that differences between the two models on this single split remain well within sampling noise.

E. Discrimination, Calibration, and Thresholds

Pooling out-of-fold predictions across repetitions yielded aggregate ROC-AUCs of 0.830 for Mixed NB, 0.828 for logistic regression, 0.731 for continuous-only Gaussian NB, and 0.812 for the decision tree (Fig. 6(a)). However, the reliability diagram in Fig. 6(b) highlights a pronounced divergence in probability calibration. While logistic regression predictions align well with true class proportions (mean CV Brier score 0.168 ± 0.026), Mixed NB produces an exaggerated sigmoidal curve with a higher Brier score (0.183 ± 0.032). This pattern is consistent with known theoretical properties of Naive Bayes under violated conditional independence: when predictors share statistical dependence, multiplying their conditional likelihoods can contribute overlapping evidence that pulls posterior probabilities toward extreme values (0 or 1) while preserving relative ranking. However, this correlation serves as a plausible contributing factor rather than a proven sole cause of probability distortion.

Table III and Fig. 7 demonstrate the empirical trade-off between sensitivity and specificity across varying decision thresholds θ. Reducing θ from 0.50 to 0.35 increases recall

TABLE II

PERFORMANCE ESTIMATES FOR THE COHORT ($N = 303$) USING 10×5-FOLD CROSS-VALIDATION AND ONE STRATIFIED HOLDOUT TEST SET ($N = 61$). CROSS-VALIDATION CELLS GIVE MEAN $\pm$ SD; BOOTSTRAP 95% INTERVALS ARE SHOWN FOR HOLDOUT ACCURACY AND AUC.

Classifier	Protocol	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC	Brier score
Mixed NB (proposed)	10×5-fold cross-validation	0.748 ± 0.048	0.733 ± 0.064	0.714 ± 0.073	0.776 ± 0.070	0.721 ± 0.055	0.831 ± 0.045	0.183 ± 0.032
Gaussian NB (numeric)	10×5-fold cross-validation	0.677 ± 0.050	0.663 ± 0.065	0.607 ± 0.082	0.736 ± 0.069	0.631 ± 0.063	0.732 ± 0.052	0.218 ± 0.027
Logistic regression	10×5-fold cross-validation	0.751 ± 0.050	0.743 ± 0.072	0.707 ± 0.077	0.788 ± 0.073	0.722 ± 0.058	0.830 ± 0.049	0.168 ± 0.026
Decision tree (fixed)	10×5-fold cross-validation	0.713 ± 0.049	0.720 ± 0.095	0.649 ± 0.124	0.767 ± 0.118	0.670 ± 0.069	0.808 ± 0.050	0.180 ± 0.028
Mixed NB (proposed)	Holdout	0.803 [0.705, 0.902]	0.767	0.821	0.788	0.793	0.891 [0.803, 0.960]	0.142
Gaussian NB (numeric)	Holdout	0.721 [0.607, 0.820]	0.720	0.643	0.788	0.679	0.765 [0.627, 0.875]	0.202
Logistic regression	Holdout	0.820 [0.721, 0.902]	0.815	0.786	0.848	0.800	0.898 [0.814, 0.966]	0.126
Decision tree (fixed)	Holdout	0.754 [0.656, 0.852]	0.810	0.607	0.879	0.694	0.860 [0.759, 0.939]	0.146

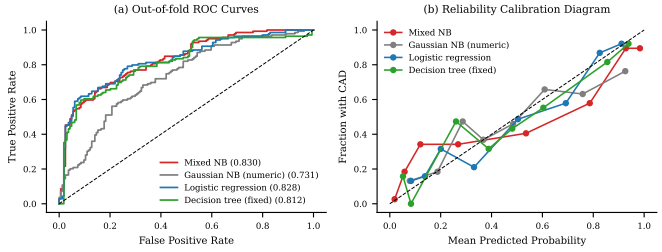


Fig. 6. (a) ROC curves and pooled AUCs from out-of-fold predictions averaged across ten repeats. (b) Reliability plots. Logistic regression has closer agreement between predicted and observed probabilities than Mixed NB.

TABLE III

MIXED NB THRESHOLD SWEEP ON AVERAGED OUT-OF-FOLD PREDICTIONS.

θ	Recall	Specificity
0.30	0.770	0.659
0.35	0.770	0.701
0.40	0.763	0.707
0.50	0.712	0.762

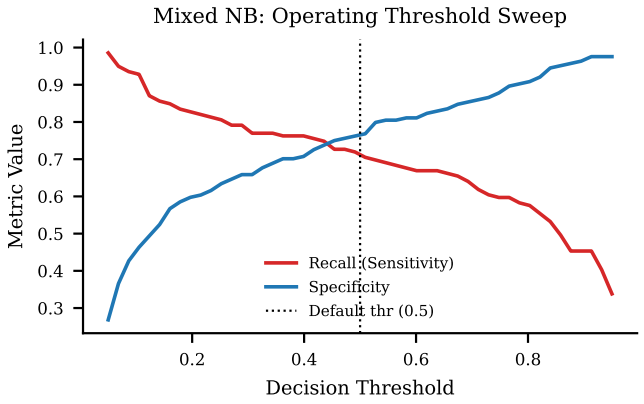


Fig. 7. Sensitivity and specificity of Mixed NB across operating thresholds. Reducing the threshold below 0.5 increases sensitivity and decreases specificity.

from 71.2% to 77.0% while reducing specificity from 76.2% to 70.1%.

VIII. DISCUSSION

The addition of chest-pain classification and exercise-induced angina provided a noticeable discriminative gain (+0.099 in mean cross-validation ROC-AUC) over numerical measurements alone. The marked prevalence of asymptomatic presentations (75.5%) and exertional angina (54.7%) in CAD cases supplies strong diagnostic information that Gaussian likelihoods on four hemodynamic variables cannot capture. Furthermore, Laplace smoothing ($\alpha = 1$) ensured robust categorical likelihood calculation without requiring ad-hoc parameter tuning on this small cohort.

At the standard 0.5 threshold, cross-validation recall for Mixed NB was 71.4% with specificity of 77.6%. On the holdout split, recall was 82.1%, missing 5 of 28 CAD cases. This variance between holdout and repeated cross-validation underscores the hazard of relying on any single favorable test partition to judge model utility.

While Mixed NB achieved ranking discrimination comparable to logistic regression (0.831 vs. 0.830), its probability calibration was visibly distorted. Violations of class-conditional independence—such as the observed inverse correlation between age and peak heart rate ($r = -0.523$ in controls)—provide a plausible contributing mechanism, as multiplying likelihoods across correlated predictors can compound redundant information and inflate posterior certainty. Nevertheless, correlation alone cannot be established as the definitive or sole cause of miscalibration in any individual prediction. Clinical decision tools that require accurate risk stratification would necessitate post-hoc recalibration techniques (e.g., Platt scaling or isotonic regression) evaluated on strictly separate data partitions.

IX. LIMITATIONS AND ETHICAL CONSIDERATIONS

1) **Historical single-site cohort.** The 303 patient records were collected at the Cleveland Clinic Foundation between 1981 and 1984. Changes in referral practices, lifestyle factors, and clinical diagnostic techniques over subsequent decades limit the direct applicability of these results to modern healthcare settings.

2) **Information loss from target binarization.** Collapsing the original categorical outcome ($\texttt{num} \in \{0, 1, 2, 3, 4\}$) into a binary target removes distinctions among the original positive diagnostic categories coded $through$.
3) **Restricted feature set.** Our model evaluates six selected variables. We excluded informative diagnostic indicators, including fluoroscopic vessel counts (`ca`) and thallium scintigraphy (`thal`), because they contained missing values.
4) **Repeated-fold dependence.** Repeated partitions of the same 303 observations share data across runs. Standard deviations across the 50 folds reflect partition variability rather than independent cohort sampling; consequently, the paired Wilcoxon test is descriptive.
5) **Single decision threshold.** The default threshold $\theta = 0.5$ balances symmetric 0-1 error. Clinical environments require setting thresholds based on asymmetric utility costs between missed ischemic diagnoses and avoidable invasive evaluations.
6) **No claim of clinical readiness.** This study evaluates statistical modeling mechanics on a historical registry. The classifiers have not undergone prospective clinical trials and must not be used for patient triage or diagnostic decision-making.

X. Conclusion

This study evaluated a hybrid Gaussian-categorical Naive Bayes classifier on the canonical UCI Cleveland cohort using leak-free parameter estimation and rigorous validation. Combining parametric Gaussian densities for continuous measurements with Laplace-smoothed categorical likelihoods for symptoms yielded an ROC-AUC of 0.831 ± 0.045, representing a 0.099 improvement over an ablated continuous-only model and matching the discrimination of regularized logistic regression (0.830 ± 0.049). However, probability calibration was less reliable (Brier score 0.183 vs. 0.168), driven by violated conditional independence assumptions in clinical measurements. Future work could explore non-parametric kernel density estimators, Bayesian networks that explicitly model conditional dependencies, and out-of-fold probability recalibration.

Reproducibility Statement

The empirical pipeline was implemented in Python 3.10 using NumPy, pandas, SciPy, scikit-learn, and Matplotlib. Fixed random seeds were assigned for the holdout split (42), cross-validation partitioning (1), and bootstrap iterations (0). Fixed hyperparameters include $\alpha = 1.0$, $\varepsilon_{\text{floor}} = 10^{-6}$, decision threshold $\theta = 0.5$, and decision tree depth 3 with minimum leaf size 5. Executing `python experiments.py` from within the `paper/` directory reads `heart_cleveland_clean.csv`, reproducing the metrics recorded in `results.json` and regenerating the figures in `figs/` (verified in the project Python environment). The LaTeX document compiles with `tectonic paper_revised.tex` (or any standard IEEEtran LaTeX distribution). All code, datasets, and configuration files are distributed within the repository.

AI Assistance Disclosure

Generative AI tools were employed as editorial assistants to assist with sentence phrasing, manuscript organization, and citation formatting. All reported numerical results, statistical tests, and plots were generated by the documented Python experimental script; AI suggestions were not treated as empirical evidence or as a substitute for verified source material. The author is responsible for verifying the dataset, code, mathematical formulations, references, and interpretations presented in this paper.

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